

CENG381 Stochastic Processes

M/M/1 and M/M/inf Queues - Question Set 1

Notation used in this question set:

M/M/1 queue:	Poisson arrivals at rate λ , single server with Exponential(μ) service times. Customers queue and are served one at a time (FCFS).
$\rho = \lambda/\mu$:	the traffic intensity. If $\rho < 1$ the queue is STABLE (a unique stationary distribution exists). If $\rho \geq 1$ the queue grows without bound.
$\pi(n)$ for M/M/1:	stationary probability of n customers in system: $\pi(n) = (1 - \rho) * \rho^n$ for $n = 0, 1, 2, \dots$
$E[N]$ for M/M/1:	mean number of customers in the system: $E[N] = \rho / (1 - \rho)$
Little's law: $E[N] = \lambda * E[T]$	where $E[T]$ is the mean time a customer spends in the system (waiting + service). Rearranging: $E[T] = E[N] / \lambda$.
M/M/inf queue:	Poisson arrivals at rate λ , INFINITE servers, each with rate μ . Every arriving customer immediately begins service (no waiting). Always stable regardless of λ and μ .
$\pi(n)$ for M/M/inf:	$\pi(n) = e^{-(\rho)} * \rho^n / n!$ i.e. Poisson(ρ) where $\rho = \lambda/\mu$. $E[N] = \rho = \lambda/\mu$ (mean service time * arrival rate).

Q1 (30 points). Customers arrive at a bank according to a Poisson process at rate $\lambda = 3$ per hour. The single teller serves customers at rate $\mu = 5$ per hour (exponential service times).

- Compute $\rho = \lambda/\mu$. Is the queue stable? Find the probability that the teller is idle (no customers in system), and the probability that there are at least 2 customers in the system.
 - Find $E[N]$, the mean number of customers in the system. Then use Little's law to find $E[T]$, the mean time a customer spends in the system (waiting + service).
 - The mean time a customer WAITS in the queue (before reaching the teller) is $E[T_q] = E[T] - 1/\mu$. Compute $E[T_q]$ and interpret: what fraction of $E[T]$ is spent waiting vs. in service?
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Q2 (35 points). Consider an M/M/1 queue with arrival rate λ and service rate μ . The stationary distribution is $\pi(n) = (1-\rho)\rho^n$ where $\rho = \lambda/\mu < 1$.

- Verify the normalisation: $\sum_{n=0}^{\infty} \pi(n) = 1$. Use the geometric series $\sum_{n=0}^{\infty} \rho^n = 1/(1-\rho)$ for $|\rho| < 1$.
 - Derive $E[N] = \rho/(1-\rho)$ by computing $\sum_{n=0}^{\infty} n \cdot \pi(n)$. Use the identity $\sum_{n=0}^{\infty} n x^n = x/(1-x)^2$ for $|x| < 1$. Show all steps.
 - The number of customers in the QUEUE (not counting the one in service) has mean $E[N_q] = \rho^2 / (1-\rho)$. Verify that $E[N_q] = E[N] - \rho$, and interpret: why is $E[N] - E[N_q] = \rho$ equal to the average number in service? (Hint: $P(\text{server is busy}) = 1 - \pi(0) = \rho$.)
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Q3 (35 points). A hospital emergency department has an unlimited number of doctors (M/M/inf model). Patients arrive at rate $\lambda = 6$ per hour and each requires Exponential(μ) service with $\mu = 2$ per hour. Since there are infinite doctors, each patient is immediately served upon arrival with no waiting.

- Find $\rho = \lambda/\mu$. Write the stationary distribution $\pi(n)$ and identify it as a named distribution. What is the probability that exactly 3 patients are in the system?
- Find $E[N]$ (mean number of patients in system) and $E[T]$ (mean time a patient spends in the system). Use Little's law: $E[T] = E[N]/\lambda$.
- Compare the M/M/inf model with an M/M/1 model with the same $\lambda = 6$ and $\mu = 12$ (one fast doctor serving at rate 12 per hour, so $\rho = 0.5$). For the M/M/1 model: compute $E[T]$. Which system gives shorter expected time in system? Which design choice is more realistic for a hospital?